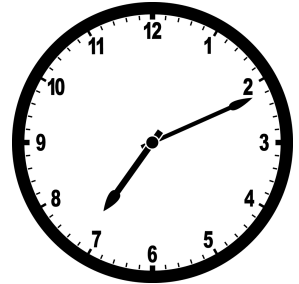


There are 9 million (9×10^6) 7-digit numbers.

There are 5040 ($7!$) 7-digit numbers using all the digits 1-7.

There are 576 ($4!$ squared) 7-digit numbers using all the digits 1-7 and divisible by 11. Call these the Seven-Elevens.

When in numerical order, we can reference the 576 Seven-Elevens by their order from least to greatest (indexed from 1 to 576.) Given an integer index, find the corresponding Seven-Eleven in the list. Get it right, and you may win a Slurpee (see Discussion)!



Input

In each dataset, you will read up to 10 integers (each from 1 to 576), one per line. 0 ends the input.

```
1
2
576
0
```

Output

Print the Seven-Eleven from the ordered list at each index requested.

```
1235476
1235674
7645231
```

Discussion

It is your choice how to find the right answer. For example: search through 9 million numbers; search through 5040 numbers; search through 576 numbers; generate the answer when you read the index; throw darts; or a method you see most reliable. (While simple, throwing darts may be the least accurate.)

Extra credit - Can you explain why " $4!$ squared"?

You won't actually win a Slurpee.

There is no extra credit.

Useful (?) math trivia: For any multiple of 11, the sum of the digits in the odd locations (first, third, ...) minus the sum of the digits in the even locations (second, fourth, ...) will equal zero or a multiple of 11. For example, in the first two answers above (which of course are multiples of 11): $(1+3+4+6) - (2+5+7) = 0$.